

Matplotlib Pie Chart, Stack Area Chart Tasks

Task 1 (Very Easy)

Create a pie chart showing **favorite fruits** of students:

- Labels: Apple, Banana, Mango, Orange
 - Values: 20, 30, 25, 25
 - Show percentage
 - Add legend at **upper right**
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Task 2

Create a pie chart of **monthly expenses**:

- Rent, Food, Travel, Savings
 - Any values of your choice
 - Explode **Savings**
 - Start angle = 180
 - Add shadow
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Task 3 (Thinking)

Create a pie chart where:

- Percentages are shown with **1 decimal**
 - One item is exploded
 - Change legend title to "**Expense Type**"
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● Area Chart Tasks

Task 4 (Very Easy)

Create an area chart showing **hours vs marks**:

- Hours: 1, 2, 3, 4, 5
 - Marks: 10, 20, 40, 60, 80
 - Line color = `red`
 - Area color = `lightgreen`
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Task 5

Create an area chart for **daily sales**:

- X-axis: Days (1 to 7)
 - Y-axis: Any sales values
 - Set transparency (`alpha`)
 - Add title and labels
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Task 6 (Thinking)

Create an area chart:

- Change line style to `dashed`
 - Change area color
 - Rotate x-axis values
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□ Stacked Area Chart Tasks

Task 7 (Very Easy)

Create a stacked area chart for:

- Days: 1, 2, 3, 4, 5
 - Sales of 3 products (any values)
 - Add legend
 - Set alpha
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Task 8

Create a stacked area chart for **3 students study hours**:

- Subjects: Math, Science, English
 - Days: 1 to 5
 - Add edge color
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Task 9 (Thinking)

Create a stacked area chart where:

- Legend is at **upper left**
 - Labels are meaningful
 - Chart shows **total contribution clearly**
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🔥 Challenge Task (Optional – Fun)

Task 10

Create **all three charts** using:

- One dataset
 - Different titles
 - Proper labels
 - Neat legend placement
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